Where Two Panoramas Can and Cannot See Depth: Epipole-Aware Fusion for Metric 3D from Sparse Panoramas

Anonymous CVPR submission

Abstract

A single panorama, passed through a modern depth network, gives a depth for every viewing direction, but the result is typically off by several percent, which is metres at street scale. Two panoramas taken a few metres apart can do much better by triangulation, with one catch: along the line joining them, the parallax vanishes and depth cannot be measured at all. For a camera moving down a street, that line points straight along the road. We derive the triangulation uncertainty of every ray in closed form, including the effect of matching noise and of errors in the relative pose. A small calibration maps it to the observed error, and we use it to fuse stereo with a learned panoramic prior in log-depth. The prior's metric scale is corrected from the well-conditioned rays, and occlusions are rejected with a mixture model. The output is metric depth with an uncertainty for each ray; where the two views carry no information, it falls back to the prior. We also outline how a dense capture of one city can serve as its own teacher, providing both supervision and calibration data. On procedural street scenes with exact ground truth, fusion lowers the median error of a simulated prior from 6.95% to 1.61%, and the predicted uncertainties are well calibrated. A simpler uncertainty derived from match confidence alone, fitted on the same data, is just as accurate in this setting. It falls behind when the confidence cannot see the geometry. With a relative-pose error of 1°/10 cm, our error is 3.39% against 3.93%. With an exposure-robust ZNCC cost, the confidence baseline reaches 5.29% where ours reaches 2.80%. All results are synthetic; an evaluation on real captures with LiDAR ground truth is still needed.

1. Introduction

Street-level panoramas cover most cities many times over. If one or two of them could be turned into metric 3D geometry close to LiDAR quality, dense city models would become much cheaper to build and to keep current. Two lines of work come close. Panoramic depth foundation models [4, 16, 18, 21] predict depth for every direction from a single image, but even the strongest report relative errors of several percent; at 20 m that is more than a metre. Generalizable multi-view methods for panoramas [2, 3, 31] build spherical cost volumes from two or more images, but they are trained to synthesize new views, not to produce calibrated metric geometry.

Two high-resolution panoramas a few metres apart contain far more metric information than either line of work uses explicitly. That information is spread very unevenly over the sphere, though. Triangulation error grows with the square of the depth, and it diverges at the *epipoles*, the two directions along the baseline. A camera on a car or a backpack moves along the street, so the baseline points down the street, and the region with no parallax is precisely the road ahead and behind (Fig. 1b). A depth estimator that ignores this has two bad options: distrust stereo everywhere, or trust it in places where it means nothing.

This paper makes that structure explicit and builds the fusion around it. Concretely, we contribute:

- A **closed-form uncertainty** for spherical two-view triangulation, for every ray, that also accounts for rotation and translation errors of the relative pose (Sec. 3.1). It agrees with Monte Carlo simulation to within 0.5%.
- A **calibration conditioned on that geometry** (Sec. 3.3). Depth, baseline, epipoles, resolution and pose error all stay in the closed form; only the photometric matching precision is learned, as a two-parameter monotone map fitted robustly on teacher data.
- **Epipole-aware Bayesian fusion** of a learned panoramic prior with one or more stereo partners in log-depth (Sec. 3.4). It corrects the prior's metric scale and rejects occlusions and moving objects with a mixture model. It returns metric depth with a per-ray uncertainty, and with a single panorama it reduces to the prior.
- A **city-as-teacher** scheme (Sec. 3.5). A dense capture of one city supplies triangulation covariances for weighting the supervision, amodal targets behind occluders that only other panoramas see, and calibration data. So far we have tested only the last of these uses.
- A **controlled study** against geometry-blind baselines fitted on the same data (Sec. 4). Without pose error, a well-calibrated match confidence does as well as the closed form. With pose error, or with an exposure-robust match-

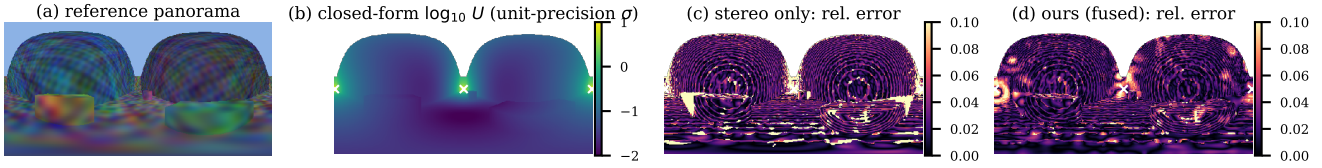


Figure 1. **Two panoramas see depth only away from the epipoles.** (a) Reference panorama of a procedural street; the second panorama was taken 4 m further down the road. (b) Our closed-form triangulation uncertainty at unit matching precision. It blows up at the two epipoles (×), which look along the street, exactly where the road continues. (c) Stereo alone fails in those regions. (d) Weighting stereo by the calibrated uncertainty and fusing it with a monocular prior keeps the stereo accuracy where it exists and falls back to the prior where it does not.

ing cost, it does not.

All experiments use procedural street scenes with exact ground truth. There are *no real-data experiments* in this draft. Section 5 describes the KITTI-360 [15] protocol with accumulated LiDAR that we consider the most important missing piece.

2. Related Work

Monocular panoramic depth. The first 360° depth networks combined equirectangular and cubemap features [10, 14, 24] or merged predictions made on perspective tangent images [21]. More recent foundation models predict metric depth for indoor and outdoor panoramas [4, 16], or lift perspective models such as Depth Anything [29] onto the sphere [18]. Perspective metric-depth models [1, 19, 30] have the same basic limitation: with one image, metric scale comes only from learned priors. We use any of these models as a prior with its own uncertainty, and correct its scale with geometry.

Multi-view panoramic reconstruction. Omnidirectional stereo with spherical sweeps goes back at least to OmniMVS [28]. Generalizable radiance-field and Gaussian-splatting methods for wide-baseline panoramas [2, 3, 31] also build spherical cost volumes, but are trained for view synthesis. Pointmap regressors [13, 25, 26] reconstruct from unposed perspective images. As far as we know, none of them models how observable depth is in each direction, and none outputs a calibrated metric uncertainty.

Triangulation uncertainty. First-order error propagation for triangulation is textbook material [5, 7], and multi-view stereo pipelines [23] use the triangulation angle as a heuristic when selecting views. We write the propagation out for spherical cameras, add relative-pose error, and use the result as the geometric backbone of a calibrated uncertainty that drives the fusion, instead of a threshold.

Uncertainty in depth estimation. Work on learned aleatoric uncertainty [11] and on uncertainty for self-supervised depth [20] is usually judged by how well the uncertainty ranks errors [9]. Calibrated regression [12] fixes the scale afterwards. Our calibration is conditioned on a closed-form geometric quantity. That quantity also carries the relative-pose error, which a learned confidence has no way to observe (Sec. 4.3).

Supervision from additional views. Behind the Scenes [27] and its distilled successor [6] train single-view density fields on KITTI-360 with extra views as supervision. Our teacher is instead a metric multi-view reconstruction of an entire city. The covariance of each teacher point weights the loss, and the same reconstruction supplies amodal targets and calibration data.

3. Method

Setting. We want the depth of a reference equirectangular panorama I_1 of width W. We are given a monocular depth prior and, optionally, one or more partner panoramas I_k. Their relative poses $(\mathbf{R}_k, \mathbf{b}_k)$ are expressed in the reference camera frame and come from GNSS/IMU or from structure from motion [22]. Each pixel defines a unit ray $\mathbf{d}$, and for each ray we estimate the metric depth z, measured as distance along $\mathbf{d}$, together with its uncertainty. All fusion is done in log-depth $\ell = \log z$, where relative errors add.

3.1. Closed-form spherical triangulation uncertainty

Take one partner at $\mathbf{b}$ and a scene point $\mathbf{X} = z\mathbf{d}$. In the triangle $(\mathbf{0}, \mathbf{b}, \mathbf{X})$, let α be the angle at the reference camera between $\mathbf{d}$ and $\mathbf{b}$, let β be the angle at the partner, and let $\gamma = \pi - \alpha - \beta$ be the parallax angle at $\mathbf{X}$ (Fig. 2). By the law of sines,

$$z = \|\mathbf{b}\| \frac{\sin \beta}{\sin \gamma}. \tag{1}$$

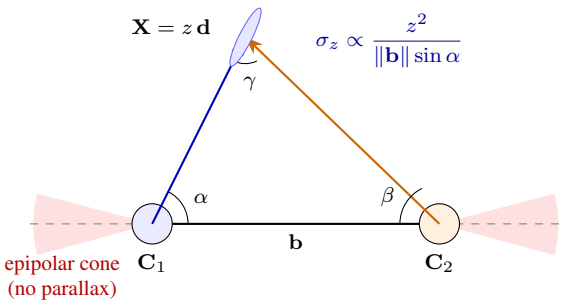


Figure 2. **Spherical two-view geometry.** The depth z along the reference ray $\mathbf{d}$ follows from the law of sines; its uncertainty (Eq. (2)) grows quadratically with depth and diverges inside the epipolar cones along the baseline, which for street-level capture is the direction of travel.

Matching noise tilts the two observed directions by small angles. Only the components inside the epipolar plane change z to first order, and differentiating gives $\partial z/\partial\alpha = \|\mathbf{b}\| \sin\beta \cos\gamma / \sin^2\gamma$ and $\partial z/\partial\beta = \|\mathbf{b}\| \sin\alpha / \sin^2\gamma$. Pose errors enter in two ways. A rotation error of the partner, with per-axis standard deviation σ_R, turns its ray inside the epipolar plane by the same angle. A position error, with per-axis standard deviation σ_t, shifts the partner ray sideways. Only the in-plane component perpendicular to the ray matters, and it moves the intersection by $\delta/\sin\gamma$. With angular matching precision σ_θ on both rays,

$$\sigma_z^2 = \frac{\|\mathbf{b}\|^2}{\sin^4\gamma}\left[\sigma_\theta^2 \sin^2\beta \cos^2\gamma + (\sigma_\theta^2 + \sigma_R^2)\sin^2\alpha\right] + \frac{\sigma_t^2}{\sin^2\gamma}. \tag{2}$$

Far from the cameras ($z \gg \|\mathbf{b}\|$, so $\beta \to \pi - \alpha$) and without pose error, this simplifies to $\sigma_z \approx \sqrt{2}\,\sigma_\theta z^2/(\|\mathbf{b}\| \sin\alpha)$. The error grows with the square of the depth, shrinks with a longer baseline, and diverges at the epipoles $\alpha \in \{0, \pi\}$.

One implementation detail matters in practice. For distant points the parallax angle is tiny, and computing $\sin\gamma$ as $\sqrt{1 - \cos^2\gamma}$ in single precision loses essentially all significant digits. We compute every sine from a cross-product norm instead. To check the derivation, we triangulated with isotropic angular noise on both rays and random pose perturbations in a Monte Carlo simulation. Equation (2) matched the simulated spread to within 0.5% in every configuration we tried (the checks are part of the released unit tests).

We write $U(\mathbf{d}) = \sigma_z/z$ for the log-depth uncertainty at *unit* matching precision, one pixel or $\sigma_\theta = 2\pi/W$, without pose error, and $P(\mathbf{d})$ for the pose-only term. U contains all of the geometry: depth, baseline length and direction, and image resolution. It depends on the unknown depth, so we evaluate it at the scale-aligned prior depth. The stereo estimate would be the wrong choice: it is least reliable exactly where U is large.

3.2. Spherical plane-sweep stereo

For every reference ray and every log-depth hypothesis, we project $z\mathbf{d}$ into the partner panorama and sample it, wrapping around in longitude. A coarse pass sweeps 64 log-spaced hypotheses between 1 and 200 m. Two refinement passes of 16 hypotheses each follow around the best match, and we take the soft-arg-min of the last pass. As matching cost we use either the box-aggregated L1 difference or $1-$ZNCC over a 5×5 window; the latter is unaffected by gain and bias changes between captures. The match confidence is the distinctiveness of the coarse cost curve, $c = (\mathrm{med}_h\, C_h - \min_h C_h)/\mathrm{med}_h\, C_h$. It drops to zero near the epipoles, where every hypothesis lands on almost the same pixel, and in textureless or occluded regions.

3.3. Geometry-conditioned calibration

Equation (2) is exact *for a given* σ_θ. The catch is that the effective matching precision is not a fixed number of pixels. It depends on the aggregation window, the texture, the cost function and the bounded depth search. On our benchmark, the implied precision ranges from about 0.1 px near the epipoles to 0.6 px perpendicular to the baseline. We therefore keep the geometry in closed form and learn only a monotone map from U to the observed error:

$$\sigma_s(\mathbf{d}) = \sqrt{\big(a(c)\,U(\mathbf{d})^\kappa\big)^2 + P(\mathbf{d})^2}, \tag{3}$$

where the exponent κ is shared and the scale $a(c)$ is piecewise constant and non-increasing in the match confidence c (six quantile bins). The pose term P has no free parameters. We fit the parameters on calibration rays with known depth, which in practice come from the city teacher of Sec. 3.5. The exponent κ is chosen by maximizing a Student-t likelihood ($\nu = 3$), which tolerates occlusions and mismatches. Each scale a is then fitted by Gaussian maximum likelihood on the t-inliers. The second step is needed because fusion requires a standard deviation, and the t-scale underestimates it by roughly 20% when errors are Gaussian. Rays deep inside the epipolar cones are left out of the fit, since the bounded depth range truncates their errors. Setting $\kappa = 1$ with a single bin recovers the pure closed form with a constant σ_θ.

3.4. Epipole-aware fusion

Scale alignment. Monocular metric depth is frequently off by a per-image scale factor. We estimate a global log-scale offset $\hat{o}$ as the precision-weighted median of $\mu_s - \mu_p$ over well-conditioned rays ($\sigma_s < 0.05$, $c > 0.5$), with weights c/σ_s^2. Its robust standard error is $s_o = 1.2533 \cdot 1.4826\,\mathrm{MAD}/\sqrt{n/25}$. The ray count n is divided by 25 because errors are correlated within each 5×5 aggregation window. We place a Gaussian prior $\mathcal{N}(0, s_p^2)$ on the scale

error of the depth prior, with $s_p = 0.1$, and apply the posterior mean $o = \hat{o}\, s_p^2/(s_p^2+s_o^2)$. When only a few reliable rays remain, or they disagree with each other, as happens under strong photometric degradation, the correction shrinks towards zero rather than locking onto a wrong scale.

Gated Bayesian fusion. In log-depth, the prior is $\mathcal{N}(\mu_p, \sigma_p^2)$ and partner k contributes $\mathcal{N}(\mu_k, \sigma_k^2)$ with confidence c_k. Occlusions, moving objects and mismatches cause gross errors. We therefore model the disagreement $\delta_k = \mu_k - \mu_p$ as a mixture of an inlier Gaussian $\mathcal{N}(0, \sigma_p^2 + \sigma_k^2)$, with prior probability π, and a uniform outlier over the log-depth range L:

$$w_k = c_k \, \frac{\pi\, \mathcal{N}(\delta_k; 0, \sigma_p^2 + \sigma_k^2)}{\pi\, \mathcal{N}(\delta_k; 0, \sigma_p^2 + \sigma_k^2) + (1-\pi)/L}. \tag{4}$$

The fused estimate is then

$$\frac{1}{\sigma^2} = \frac{1}{\sigma_p^2} + \sum_k \frac{w_k}{\sigma_k^2}, \qquad \mu = \sigma^2 \Big(\frac{\mu_p}{\sigma_p^2} + \sum_k \frac{w_k\, \mu_k}{\sigma_k^2} \Big). \tag{5}$$

Near the epipoles $\sigma_k \to \infty$, and the result is simply the scale-aligned prior. Perpendicular to the baseline, stereo dominates. With a single panorama there are no partners, and the output is the prior. Every step is differentiable, so the module can also sit inside a learned network (Sec. 3.5).

3.5. City as teacher

Suppose one city has been captured densely, with many panoramas per street. We reconstruct it once with multi-view geometry, using SfM and MVS [22, 23] or surface splatting [8]. A teacher point $\mathbf{X}$ seen from centres $\mathbf{C}_i$ at distances r_i, along unit directions $\mathbf{u}_i$, has triangulation information $\mathbf{\Lambda} = \sum_i (\mathbf{I} - \mathbf{u}_i \mathbf{u}_i^\top)/(r_i \sigma_\theta)^2$ and covariance $\mathbf{\Sigma} = \mathbf{\Lambda}^{-1}$. We invert in double precision, because points seen from a single view give nearly singular matrices. For a student panorama we render targets in two layers. The first is the nearest teacher surface along each ray. The second is the first surface behind it, for example behind a parked car or around a corner, which only other panoramas of the city observe. Each target carries its variance $\mathbf{d}^\top \mathbf{\Sigma} \mathbf{d}$ along the student ray. A student that predicts (μ, σ) in log-depth is trained with a Gaussian negative log-likelihood of variance $\sigma^2 + \mathbf{d}^\top \mathbf{\Sigma} \mathbf{d}/z^2$. Poorly triangulated teacher points therefore supervise weakly instead of wrongly. The same teacher also provides the calibration rays for Eq. (3). Our reference student uses one equirectangular encoder, with wrap-around padding in longitude, that feeds five components: a prior head, a stereo cost volume on learned features, the closed-form uncertainty with a learned global scale, the fusion above, and an amodal second-layer head.

4. Experiments

4.1. Setup

Benchmark. We do not currently have real street panoramas with dense metric ground truth (Sec. 5). Instead we generate procedural street canyons with exact depth. Each scene has a ground plane, two building façades 5–9 m from the camera with random heights, end walls 60–120 m away, 3–7 box-shaped parked vehicles, and sky. Every surface carries a view-independent, multi-scale procedural texture. The reference panorama sits at the origin and the partner b metres further along the street. This is the hardest placement, because both epipoles then look down the street. Unless noted otherwise, panoramas are 256×512 and $b = 4$ m. We simulate the monocular prior as the true log-depth plus spatially smooth noise with a standard deviation of 6%, plus a global scale error drawn uniformly from $\pm 10\%$; the prior reports a constant $\sigma_p = 0.06$. This is in line with the accuracy reported for current panoramic foundation models [16]. Calibration uses six scenes (seeds ≥ 1000) that are disjoint from the ten test scenes, and ground truth stands in for the city teacher.

Baselines. All fusion variants share the same stereo, prior, scale alignment and gate. They differ only in how the stereo uncertainty σ_s is set. *Constant* uses $\sigma_s = s$ and *confidence* uses $\sigma_s = s/c$; in both, s is fitted on the same calibration rays with the same robust estimator as ours. *Selection* keeps, for each ray, whichever source has the smaller calibrated σ. We also run our closed form without calibration ($\sigma_\theta = 0.5$ px), and report the prior alone, the scale-aligned prior and stereo alone.

Metrics. We report the median relative depth error, overall and binned by the angle between ray and baseline, and the fraction of rays within 20 cm. For the uncertainty we report the fraction of errors inside the predicted 1σ (ideally 0.683), the Gaussian negative log-likelihood (NLL) in log-depth, which is a proper scoring rule, and the area under the sparsification error curve (AUSE) [9].

4.2. Main results

Table 1 and Fig. 3 show the pattern we expected. Stereo alone is accurate perpendicular to the baseline (1.49% at 60–90°) and useless near the epipoles (21%). The prior is equally mediocre everywhere: 6.95% before scale alignment and 3.94% after it. Fusion tracks the better of the two, bringing the median error of the raw prior down from 6.95% to 1.61%, and 0.71 of its errors fall inside the predicted 1σ. Scale alignment is important. Without it, the error near the epipoles rises from 4.30% to 6.51%. Once the precision is calibrated, evaluating U at the stereo depth instead of the prior depth makes no difference. The uncalibrated closed

Method	Median rel. error (%) ↓, by angle to baseline					<20 cm (%) ↑	Cal. 1σ	NLL ↓	AUSE ↓
	0–10°	10–30°	30–60°	60–90°	all	all	(0.68)	all	all
Prior only	6.81	7.04	6.99	6.92	6.95	29.80	0.44	-0.73	0.481
Prior + scale alignment	4.45	4.03	3.83	3.98	3.94	47.12	0.69	-1.40	0.534
Stereo only	21.07	3.03	2.05	1.49	1.78	68.99	**0.68**	26.04	0.386
Fusion, constant σ_s	4.90	2.41	1.76	1.48	1.64	73.66	0.73	-2.17	**0.329**
Fusion, confidence σ_s	4.49	**2.35**	**1.73**	1.43	**1.60**	**74.10**	0.73	**-2.22**	0.331
Selection (min. σ)	4.45	3.07	2.03	1.47	1.74	70.60	0.70	-2.03	0.354
Ours, closed form (σ_{px}=0.5)	4.38	3.03	1.91	1.43	1.67	72.17	0.63	-1.36	0.461
Ours (calibrated)	**4.30**	2.63	1.77	**1.41**	1.61	73.63	0.71	-2.16	0.361

Table 1. **Main results** (256×512, $b = 4$ m along the street, 10 test scenes). The calibrated closed form is best or tied in accuracy near the epipoles and perpendicular to the baseline, and its uncertainties are well calibrated. A confidence-based σ_s fitted on the same calibration data is equally accurate in this setting (see text).

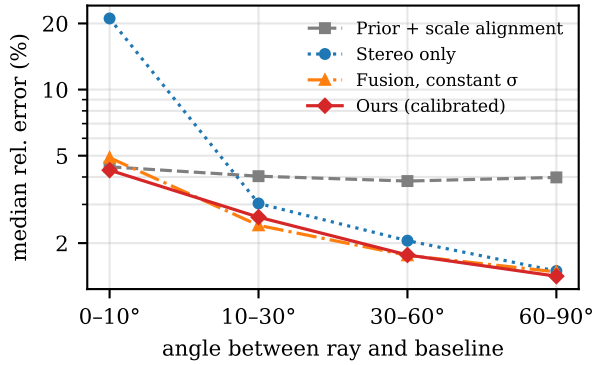


Figure 3. Median relative error against the angle between ray and baseline. Stereo diverges near the epipoles and the scale-aligned prior is flat; fusion tracks the better of the two.

form ($\sigma_\theta = 0.5$ px) is as accurate overall but poorly calibrated (NLL -1.36 against -2.16). This is why we added Eq. (3): on this benchmark, the implied matching precision changes by a factor of six with the angle to the baseline.

A negative result. Fitted on the same calibration rays, the geometry-blind *confidence* baseline is as accurate as our method (1.60% vs. 1.61%), and its NLL is slightly better. In hindsight this makes sense. The match confidence already collapses near the epipoles, because all hypotheses project to almost the same pixel there, so it encodes much of the observability implicitly. The picture is the same when the calibration is transferred to other baselines and resolutions (Tab. 2): all calibrated fusion variants stay within 0.1 percentage points of each other. Without pose error, then, the closed form does *not* improve accuracy over a well-calibrated confidence on this benchmark.

4.3. Where geometry matters: relative-pose error

Poses from GNSS/IMU or SfM are never exact, and their errors do not affect all rays alike. By Eq. (2), a translation error moves the depth by $\delta / \sin\gamma$. That is negligible for nearby points perpendicular to the baseline and dominant for distant points and near the epipoles. The match confidence cannot see any of this: the cost curve can be sharply peaked at the wrong depth. In Tab. 3 we perturb the partner pose. For small errors (0.1°, 2 cm) every method behaves the same. At 0.3°/5 cm the geometry-blind uncertainties become overconfident, with only 0.59–0.60 of errors inside 1σ. Ours stays calibrated (0.70) and has the best NLL, at equal accuracy. At 1°/10 cm the pose term changes the outcome. Our median error is 3.39% against 3.93% for the best baseline, the fraction inside 1σ rises from 0.46 to 0.67, and the NLL improves from -0.82 to -1.54.

Recalibrating the geometry-blind baselines under the same pose noise does restore calibration at moderate noise. At large noise it backfires: it inflates σ_s for every ray, no ray stays reliable enough for scale alignment, and the result drifts towards the unaligned prior (6.5–6.8%). Our model still trusts nearby rays perpendicular to the baseline, where the pose term stays small. We do assume the noise level is known; in practice it would come from the SfM covariance or the GNSS/IMU specification.

Partners with different baselines. With two partners, at +2 m and −8 m, all calibrated variants reach the same accuracy (1.19–1.21%) and similar calibration. On this benchmark we see no benefit from geometric weighting across partners.

4.4. Capture effects

Table 4 varies the capture conditions, and two things are worth pointing out. The first is that the cost function matters. L1 works best on clean images and with sensor noise,

	b=2 m		b=8 m		128×256		512×1024	
Method	Med. (%)	NLL	Med. (%)	NLL	Med. (%)	NLL	Med. (%)	NLL
Prior + scale alignment	3.92	-1.39	3.91	-1.39	3.99	-1.37	3.97	-1.39
Stereo only	1.65	2.02	2.16	133.00	2.21	5.59	1.49	75.30
Fusion, constant σ_s	1.57	-2.25	1.82	-1.87	1.94	-2.03	1.46	-2.18
Fusion, confidence σ_s	1.52	-2.32	1.81	-1.92	1.89	-2.10	1.41	-2.22
Ours (calibrated)	1.51	-2.29	1.82	-1.89	1.97	-2.03	1.38	-2.00

Table 2. **Transfer without recalibration.** All methods are calibrated at $b = 4$ m and 256×512 and evaluated at other baselines and resolutions (5 scenes; 3 at 512×1024).

	0.1°/2cm			0.3°/5cm			1°/10cm		
Method	Med.	Cal.	NLL	Med.	Cal.	NLL	Med.	Cal.	NLL
Constant σ_s	1.77	0.70	-2.08	2.34	0.59	-1.70	3.99	0.44	-0.51
Constant σ_s, refit	1.79	0.73	-2.09	2.41	0.71	-1.83	6.77	0.44	-0.77
Confidence σ_s	**1.73**	0.70	-2.13	2.30	0.60	-1.79	3.93	0.46	-0.68
Confidence σ_s, refit	1.74	0.72	**-2.13**	2.36	0.71	-1.87	6.52	0.44	-0.82
Ours w/o pose term	1.74	**0.69**	-2.08	**2.26**	0.60	-1.78	3.86	0.46	-0.68
Ours	1.75	0.71	-2.09	2.28	**0.70**	**-1.89**	**3.39**	**0.67**	**-1.54**

Table 3. **Relative-pose error.** The partner pose given to every method is perturbed by rotation and translation noise with the stated per-axis standard deviations; the images themselves are rendered with the true poses. "Refit" means the geometry-blind baseline was recalibrated on calibration scenes with the same pose noise. Ours plugs the known noise level into the pose term of Eq. (2) and is *not* recalibrated. 8 test scenes.

Method	clean	exposure	noise	moving cars	all
Confidence σ_s, L1	**1.59**	5.69	2.26	**1.62**	9.07
Confidence σ_s, ZNCC	5.29	5.42	6.50	5.32	6.72
Ours, L1	1.60	5.96	**2.24**	1.63	8.51
Ours, ZNCC, w/o outlier gate	3.06	3.06	3.95	3.09	4.57
Ours, ZNCC	2.80	**2.88**	3.92	2.82	**4.56**

Table 4. **Capture effects** (median relative error, %, 8 scenes). We apply an exposure change to the partner (gain 0.6, bias 0.15), add sensor noise ($\sigma = 0.03$), move half of the parked cars by 3 m between captures, and combine all three. Each cost function has its own calibration.

Calibration data	κ	Med. rel. (%)	Cal. 1σ	NLL
Ground truth (oracle)	0.38	1.61	0.71	-2.16
Teacher, naive	0.40	1.62	0.73	-2.15
Teacher, $\geq$2 views	0.35	1.62	0.73	-2.16
Teacher, covariance-aware (ours)	0.30	1.61	0.73	-2.17

Table 5. **Calibration data.** The precision model of Eq. (3) is fitted on depth from a simulated city teacher instead of ground truth: 7 panoramas per street with lateral offsets, each point displaced by a draw from its own triangulation covariance, and 0.5 px matching at 2048 px width. Evaluation as in Tab. 1.

but ZNCC copes far better with an exposure change between captures (2.88% against 5.96%) and with all effects combined (4.56% against 8.51%). The second is that the confidence baseline falls apart once ZNCC is used: 5.29% against our 2.80% on clean data, and 6.72% against 4.56% in the combined setting. With L1, the confidence happens to encode the epipolar degeneracy; with ZNCC it does not, and nothing guarantees that it will for an arbitrary cost. The closed form does not rely on that. The outlier gate helps a little on clean data and with moving cars (from 3.06% to 2.80% and from 3.09% to 2.82%), and not at all in the combined setting.

4.5. Calibrating from a noisy city teacher

Fitting the calibration on noisy teacher depth instead of ground truth costs nothing we can measure (Tab. 5). Every variant reaches 1.61–1.62% with calibrated uncertainties. If the teacher is treated as exact, the fitted scales come out slightly too large, because teacher error is absorbed into the matching precision; adding the teacher covariance in quadrature removes that bias. Downstream, though, the difference is negligible here. The simulated teacher is about $8\times$ more precise than the 512 px stereo it calibrates. Covariance-aware calibration should only matter when teacher and stereo have comparable precision, and we have not tested that case.

5. Limitations and Future Work

No real data yet. Every result in this draft comes from procedural scenes with a simulated prior. Real panoramas bring non-Lambertian surfaces, vegetation, sky, rolling-shutter and stitching artefacts, and a prior whose errors have spatial structure that our simulation does not reproduce. The experiment that will decide the matter is KITTI-360 [15]. Its two 180° fisheye cameras (MEI model [17]) cover nearly the full sphere and can be resampled to equirectangular images, and its accumulated LiDAR provides metric ground truth, including in occluded regions. We plan to use pairs of frames 2–8 m apart along the trajectory, priors from Depth Any Panoramas [16] and DA360 [4], calibration on training sequences, and evaluation on held-out sequences. The comparison will include Splatter-360 [3], PanSplat [31], and pointmap regressors [13, 25] run on cube faces.

The learned student is untested. The city-as-teacher training of Sec. 3.5 is implemented. A small reference student trained for 400 steps on a CPU did not converge, and its errors stayed far above those of the prior, so we report no student results. The teacher has been tested only as a source of calibration data (Sec. 4.5). Whether amodal supervision and covariance-weighted teacher losses help a foundation-model student will require training on GPUs.

Modelling assumptions. Partners share the reference image, so treating several of them as independent overstates the precision. Moving objects are rejected as outliers, and in those regions the depth comes from the prior. Relative-pose error enters σ_s only through a per-axis standard deviation. When poses come from SfM, the full pose covariance should be propagated instead.

6. Conclusion

Two panoramas hold precise metric information about a scene, but only away from their epipoles. A closed-form, pose-aware triangulation uncertainty, with a small calibration of the matching precision, makes this explicit. Fused on that basis with a monocular prior, it yields metric depth with a per-ray uncertainty, and falls back to the prior where the geometry says nothing. In our controlled experiments it cuts the error of a simulated foundation prior by more than $4\times$. Without pose error, a well-calibrated match confidence does just as well. The closed form is clearly better once the relative pose is uncertain, which is the usual situation for street-level capture, and when an exposure-robust cost is used, because the match confidence then no longer encodes the epipoles. Whether all this brings real street-level panoramas close to LiDAR accuracy is the next question to answer.

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